

Algebraic Geometry – Exercise Sheet 4

Week 4: Reduced, irreducible and Noetherian schemes; dimension and projective examples

One Monday exercise session, followed by a revision bank for the final exam

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — reduced, irreducible and connected schemes

Exercise 1. Irreducible but non-reduced

★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(x^2, xy), \quad X = \operatorname{Spec}(A).$$

- (i) Show that every prime ideal of A contains the class of x and identify X_{red} .
- (ii) Is X reduced? Is it irreducible? Is it connected?
- (iii) Identify the generic point of X .

Hint 1 (conceptual). Nilpotent elements belong to every prime ideal. Reduction does not change the underlying topological space.

Hint 2 (technical). The radical of (x^2, xy) in $\mathbb{k}[x, y]$ is (x) .

Exercise 2. A parabola and a disjoint point

★★ **CORE**

In $\mathbb{A}_{\mathbb{k}}^2 = \operatorname{Spec}(\mathbb{k}[x, y])$, let

$$C = \mathbb{V}(y - x^2), \quad P = \mathbb{V}(x, y - 1), \quad Z = C \cup P.$$

- (i) Show that

$$(y - x^2) + (x, y - 1) = \mathbb{k}[x, y].$$

Deduce that C and P are disjoint.

- (ii) Determine generators for the ideal of Z . You may use that for comaximal ideals I, J one has $I \cap J = IJ$.
- (iii) Determine the irreducible components of Z . Is Z connected? Is it reduced?
- (iv) Compute $\Gamma(\mathcal{O}_Z)$.

Hint 1 (technical). For (i), write 1 using $y - x^2$, x , and $y - 1$.

Hint 2 (conceptual). The ideal of a union of closed subschemes is the intersection of their ideals. For the last question, use the Chinese remainder theorem.

Exercise 3. Connected as a scheme, disjoint over $\mathbb{R}$

★★ **CHOICE**

In $\mathbb{A}_{\mathbb{R}}^2$, set

$$L = \mathbb{V}(x), \quad C = \mathbb{V}(x - y^2 - 1), \quad Z = L \cup C.$$

- (i) Show that L and C are irreducible and compute their function fields.
- (ii) Show that their sets of $\mathbb{R}$ -rational points are disjoint.
- (iii) Compute $L \cap C$ as a scheme. Deduce that Z is connected.
- (iv) Show that Z is reduced.

Hint 1 (conceptual). The intersection of two affine closed subschemes is obtained by adding their ideals. A closed point of an $\mathbb{R}$ -scheme need not have residue field $\mathbb{R}$.

Hint 2 (technical). The quotient by $(x, x - y^2 - 1)$ is $\mathbb{R}[y]/(y^2 + 1)$.

Exercise 4. The function field of the cusp

★★ **CORE**

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and compute its function field.

Suggested strategy. Use the parametrisation $x = t^2$, $y = t^3$, first to embed the coordinate ring in $\mathbb{k}[t]$, and then to compare fraction fields.

Hint 1 (technical). Every class modulo $y^2 - x^3$ can be represented as $A(x) + yB(x)$. After substituting $x = t^2$, $y = t^3$, compare even and odd powers of t .

Noetherian schemes and dimension

Exercise 5. Why algebraic schemes are Noetherian

★★ CORE

Let X be an algebraic $\mathbb{k}$ -scheme.

- (i) Explain why X admits a finite affine open covering

$$X = U_1 \cup \cdots \cup U_r, \quad U_i = \operatorname{Spec}(A_i),$$

with every A_i a finitely generated $\mathbb{k}$ -algebra.

- (ii) Show that every U_i is a Noetherian topological space.
 (iii) Deduce that X is Noetherian.
 (iv) Deduce that X has only finitely many irreducible components.

Hint 1 (conceptual). Use Hilbert's basis theorem for the affine pieces. For (iii), an ascending chain of open subsets stabilises if it stabilises after restriction to every member of a finite open cover.

Exercise 6. Components, generic points and dimension

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that

$$I = (x) \cap (y, z).$$

- (ii) Determine the irreducible components and their generic points.
 (iii) Compute the dimension and the function field of each component.
 (iv) Is X reduced? connected? irreducible? integral?

Hint 1 (technical). If $f = xg$ belongs to (y, z) , reduce modulo (y, z) and use that $\mathbb{k}[x]$ is a domain.

Exercise 7. Dimensions revisited

★ CHOICE

Use the dimension results for algebraic $\mathbb{k}$ -schemes to compute the following.

- (i) The dimension of the cusp in Exercise 4.
 (ii) The dimension of the scheme Z in Exercise 2.
 (iii) The dimension of $\mathbb{P}_{\mathbb{k}}^n$.

Hint 1 (technical). For an integral algebraic $\mathbb{k}$ -scheme, use the transcendence degree of its function field. For a reducible Noetherian scheme, take the maximum of the dimensions of its irreducible components. For projective space, use the standard affine cover together with a chain of linear subspaces.

Projective schemes and final revision bank

Exercise 8. Closed and non-closed points of $\mathbb{P}_{\mathbb{F}_3}^2$

★★ CORE

Let

$$X = \mathbb{P}_{\mathbb{F}_3}^2.$$

- (i) Show that if $p \in X$ is a closed point, then

$$|\kappa(p)| = 3^m$$

for some integer $m \geq 1$.

(ii) Show that for every $m \geq 1$ there exists a closed point $p \in X$ with

$$|\kappa(p)| = 3^m.$$

(iii) How many points $p \in X$ satisfy $|\kappa(p)| = 3$?

(iv) Show that if $p \in X$ is not closed, then $\kappa(p)$ is infinite and

$$\text{trdeg}_{\mathbb{F}_3} \kappa(p) \leq 2.$$

Hint 1 (conceptual). Work in a standard affine chart of $\mathbb{P}_{\mathbb{F}_3}^2$ and interpret a point as a prime ideal of a polynomial ring in two variables.

Exercise 9. A projective scheme with a non-standard grading

★★ **CORE**

Let

$$S = \mathbb{k}[x, y], \quad \deg(x) = 1, \quad \deg(y) = 2, \quad X = \text{Proj}(S).$$

(i) Show that $D_+(x)$ and $D_+(y)$ cover X .

(ii) Compute $(S_x)_0$ and $(S_y)_0$.

(iii) Describe the overlap and deduce that $X \simeq \mathbb{P}_{\mathbb{k}}^1$.

Hint 1 (technical). On $D_+(x)$ use $u = y/x^2$; on $D_+(y)$ use $v = x^2/y$.

Exercise 10. The projective conic

★★ **CHOICE**

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S),$$

with the standard grading.

(i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.

(ii) Show that $D_+(x_0)$ and $D_+(x_2)$ cover C , and describe their overlap.

(iii) Show that

$$\mathbb{P}_{\mathbb{k}}^1 \longrightarrow C, \quad [s : t] \longmapsto [s^2 : st : t^2]$$

is an isomorphism.

Hint 1 (technical). On $D_+(x_0)$ use $u = x_1/x_0$; on $D_+(x_2)$ use $v = x_1/x_2$.

Exercise 11. Three charts of a weighted projective plane

★★★ **CORE**

Let

$$S = \mathbb{k}[x, y, z], \quad \deg(x) = 1, \quad \deg(y) = 2, \quad \deg(z) = 3, \quad X = \text{Proj}(S).$$

Set

$$U_x = D_+(x), \quad U_y = D_+(y), \quad U_z = D_+(z).$$

(i) Show that U_x, U_y, U_z cover X and compute their coordinate rings.

(ii) Describe explicitly the transition isomorphisms on the three pairwise overlaps.

(iii) Verify the cocycle condition on $U_x \cap U_y \cap U_z$.

Hint 1 (technical). Useful degree-zero elements are

$$u = \frac{y}{x^2}, \quad v = \frac{z}{x^3}; \quad a = \frac{x^2}{y}, \quad b = \frac{xz}{y^2}, \quad c = \frac{z^2}{y^3};$$

and

$$r = \frac{x^3}{z}, \quad s = \frac{xy}{z}, \quad w = \frac{y^3}{z^2}.$$

Exercise 12. Removing generic points and their closures in $\mathbb{P}^3$

★★ **CHOICE**

Let

$$X = \mathbb{P}_{\mathbb{k}}^3$$

and, for $0 \leq m \leq 2$, let

$$L_m = \mathbb{V}(x_{m+1}, \dots, x_3) \simeq \mathbb{P}_{\mathbb{k}}^m.$$

Let η_m be the generic point of L_m .

- (i) Identify η_m as a homogeneous prime ideal and compute $\overline{\{\eta_m\}}$.
- (ii) Consider the ringed subspace

$$X \setminus \{\eta_m\}.$$

For which values of m is it a scheme? When it is a scheme, is it affine?

- (iii) Consider instead

$$X \setminus \overline{\{\eta_m\}} = X \setminus L_m.$$

Show that this is always a scheme. For which values of m is it affine?

Hint 1 (conceptual). Recall the criterion for removing a single point with the induced ringed-space structure. For the last question, distinguish a hyperplane from a linear subspace of codimension at least two.